Sahaj Bhadja

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Summary

Software engineer building **Python/React applications**, distributed systems, Docker/Kubernetes deployments, and AI workflows; delivered secure platforms handling 1M+ transactions and 10K+ daily requests.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected a **multi-tenant data platform** with normalized relational schemas and row-level isolation for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through indexed query and access-pattern optimization.
- Built **ETL-style integrations** across FastAPI, PostgreSQL, SQL Server, and cloud storage, processing 5,000+ monthly jobs with incremental loads, reconciliation, validation, error handling, and CI/CD releases achieving 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **full-stack React and Flask features** for an AI grading platform handling 10,000+ daily requests, translating instructor workflows into secure upload, dashboard, and feedback experiences for 150+ users.
- Integrated **OpenAI, Anthropic Claude, and Qwen providers** behind switchable APIs, adding OCR fallbacks, usage logging, cost tracking, and concurrent processing while reducing feedback turnaround time by 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **Python data and evaluation pipelines** processing 100K+ trajectory records, creating reproducible test scenarios and diagnostics that improved long-horizon planning 18% and strengthened analysis of complex model behavior.
- Documented **model constraints, benchmark results, and failure patterns** in technical reports, translating complex experimental behavior into actionable engineering guidance that improved closed-loop planning stability by 15% across simulations.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Optimized **distributed API and ETL integrations** using OOP, AWS Glue, S3, SQL, indexing, and caching, improving response times 30%, pipeline efficiency 68%, and storage costs 15% across production workloads.
- Shipped **cross-platform software through Agile sprints** with Microsoft and GitHub stakeholders, maintaining 20+ unit/integration tests and Jenkins CI/CD across Windows, Linux, and macOS release environments with reliable delivery.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **React and Node.js features across eight microservices** for authentication, customer records, billing, payments, permissions, and notifications, synchronizing 2M+ records through WebSockets and Azure SQL across locations.
- Automated **Docker packaging, testing, and staged deployments** through Jenkins CI/CD, cutting release time 50% while improving consistency across development environments, code reviews, and reliable production rollout workflows.

Skills

Languages: Python, JavaScript, TypeScript, Java, SQL, C++, Bash
Frontend: React.js, Next.js, TailwindCSS, HTML, CSS, responsive interfaces
Backend: FastAPI, Flask, Node.js, Express.js, REST APIs, microservices, distributed systems, LLM systems
Cloud & Delivery: GCP, AWS, Azure, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git
Data & Security: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, ETL, validation, indexing, RLS, RBAC

Projects

FinNexus Live | *Next.js, TypeScript, MongoDB, Kubernetes*
- Built a **Kubernetes-deployed fintech platform** supporting 50,000+ users and 1M+ transactions, combining secure bank linking, real-time updates, transaction filtering, MongoDB-backed access, and CI/CD automation for scalable operations.

Big-Brain Live | *GCP, LLaMA-2, FAISS, Pinecone, PostgreSQL*
- Built a **GCP-backed LLM document platform** with FAISS, Pinecone, PostgreSQL, and REST APIs, indexing 50K+ records through incremental updates while reducing retrieval latency 40% for scalable semantic search.

HMIS Live | *TypeScript, MERN, ABDM APIs, Sentry*
- Built a **secure information platform** with consent-based record access, appointment workflows, TypeScript dashboards, ABDM API integrations, Twilio notifications, and Sentry monitoring, reducing administrative workload by 30% for staff.